

TNRD V6: zstd compression levels

Track N Race · TNRD V6 writer · udp_capture_to_tnrd · measured 9 October 2026

Summary

- **Keep level 9.** It is the default, and nothing above it is worth the cost. The smallest file any level made is 3.7% smaller (level 22), for 7.3× the time.
- **Levels 10–14 change nothing.** They are within 0.1% of level 9 for all drivers, except level 13, which is 2.2% *bigger*.
- **The first real step is level 15:** 2.2% smaller for 1.4× the time. Level 18 gets 3.3% for 2.9×. Levels 20–22 level off at 3.6–3.7% for 5–7×.
- **Below 9, files grow and nothing gets faster.** Levels 1–8 make files up to 4.8% bigger, and their conversion times sit in the same band as level 9. At these levels compression is small next to parsing the capture.
- **Driver-only files follow the same curve** at 1/16 of the size: 4.74–5.18 MB against 73.6–80.1 MB.
- **Opening a file and peak memory don't depend on the level.**

Windows 11, AMD Ryzen 7 7435HS (8 cores, 16 threads), 24 GB. Data: the 92-minute São Paulo race, as the UDP capture `udp_capture_sao_paulo.bin` (2.5 GB, 2,498,862 datagrams). Each level was converted once with `udp_capture_to_tnrd --scope both <capture> <dir> <level>`, one level at a time. `--scope both` records through the same in-memory session the desktop apps use, and writes an **all-drivers** and a **driver-only** file in the same run. **Conversion time** is wall time for the whole capture, parsing included. **Opening** is the reader loading the file (`--validate`). Harness:
`tools/udp_capture_to_tnrd/zstd_sweep.ps1; data: docs/compression-reports/zstd-levels.csv.`

1. File size

Every level, 1 to 22, as a change against level 9. Level 9 made a 76.44 MB all-drivers file and a 4.92 MB driver-only file.

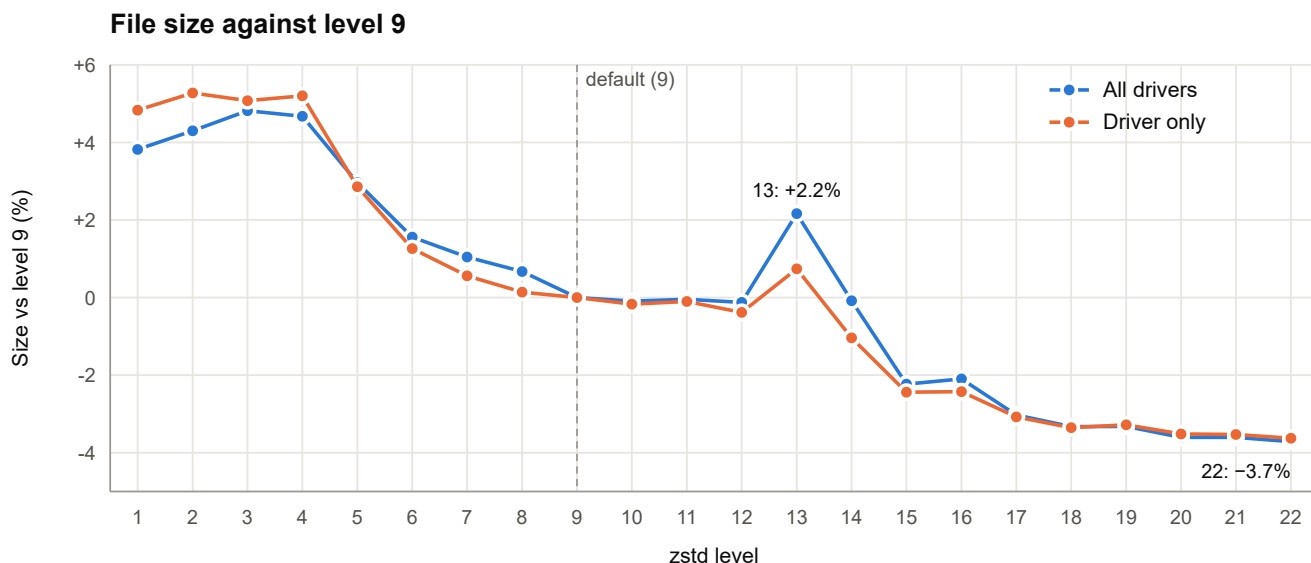


Figure 1. File size against level 9. Lower is smaller. Both files are on the same percentage scale.

Across all 22 levels the all-drivers file spans 73.60 to 80.12 MB, under 9% from the largest to the smallest. Sizes don't fall steadily with the level. Levels 2–4 are larger than level 1, level 13 is larger than 12 and 14, and level 16 is a little larger than 15.

2. Conversion time

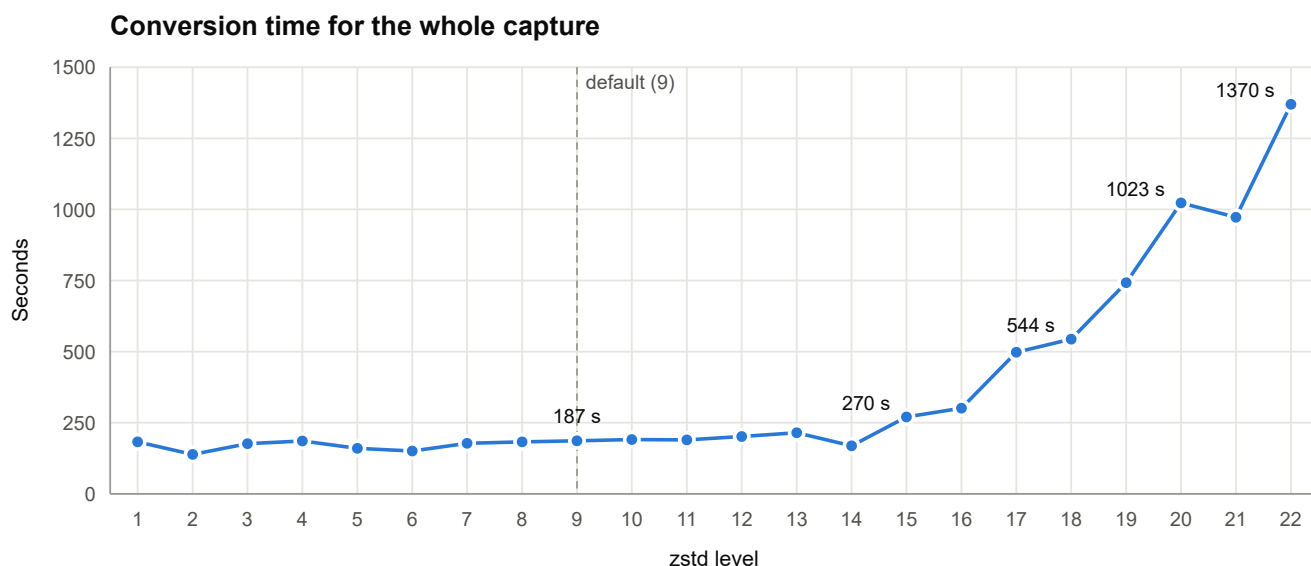


Figure 2. Wall time to convert the whole 2.5 GB capture at each level, parsing included. One run per level.

Levels 1–14 all took 139–215 s. Each level ran once, and the spread there is mostly noise: level 2 (139 s) and level 14 (169 s) were faster than level 9 (187 s). From level 15 the time climbs: 270 s at 15, 544 s at 18, 1023 s at 20 and 1370 s at 22. CPU time was 15–35 s above wall time at every level. That's the writer thread working alongside the parser.

Level	Extra time over level 9	All drivers saved	Driver only saved
15	+83 s	1.70 MB	0.12 MB
18	+357 s	2.54 MB	0.17 MB
20	+836 s	2.76 MB	0.17 MB
22	+1183 s	2.84 MB	0.18 MB

3. Size against time

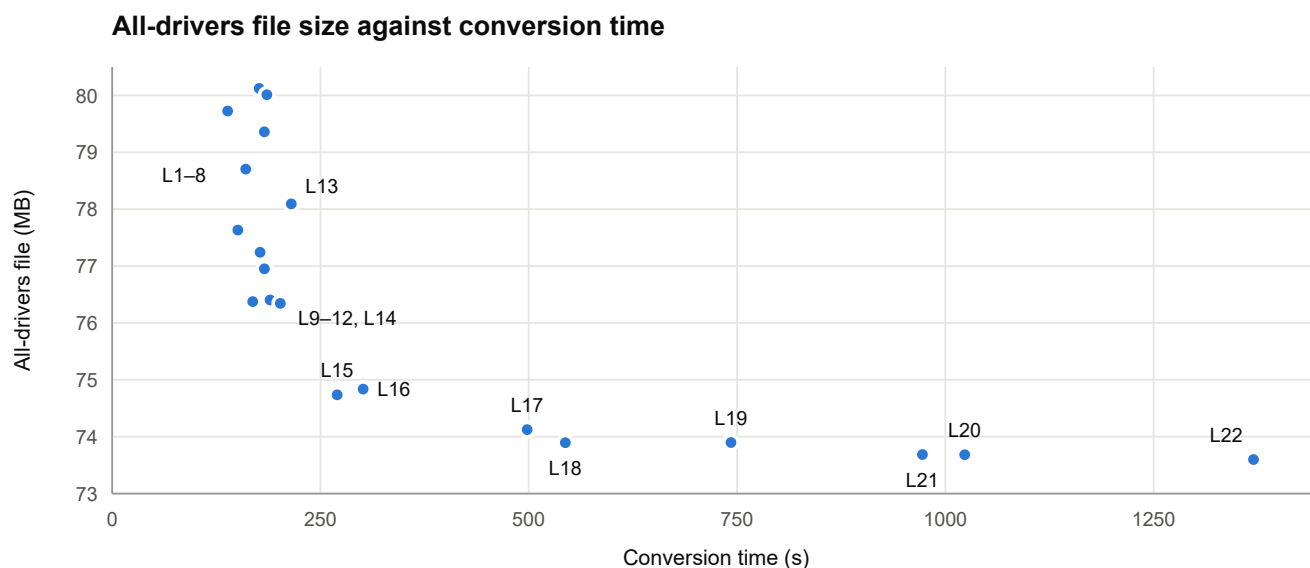


Figure 3. All-drivers file size against conversion time, one dot per level. Levels 1–8 form the upper column; 9–12 and 14 overlap at about 76.4 MB.

Levels 1–14 sit in one column: their times overlap, and only the size changes. Past that, each extra 1% costs several minutes per race. The levels worth choosing are 9, 15, 18 and 20. Every other level is either no smaller than one of these or slower for the same size.

4. Opening the file and memory

Measure	Levels 1–22
Opening an all-drivers file	1.33–1.79 s
Opening a driver-only file	1.07–1.43 s
Converter peak working set	343–396 MB

Opening shows no trend with the level. The two highest readings, at level 19 (1.79 s and 1.43 s), sit between ordinary readings at 18 and 20. Reading zstd costs about the same whatever level wrote it, so the reader doesn't care which level is chosen. Peak memory doesn't change either.

5. Why the curve flattens early

A V6 file is not one compressed stream. Every chunk is compressed on its own, one data type for one car over one lap (`compressFrame` in `TNRD_V6.cpp`). Shared records (session, participants, events) are each compressed against the previous record of their type. These inputs are small. The higher levels mostly add a bigger search window and a deeper match search, and inside a single small chunk there is little left for them to find. The gain stops at about 3.7%.

6. What's proven and what isn't

Claim	Status
Sizes and times for levels 1–22, both files	Measured (one run per level, 9 October)
Levels above 9 save at most 3.7%	Measured on one race
Levels 1–14 take the same time within run-to-run noise	Measured; noise of about ± 25 s shown by levels 2 and 14
Opening time and peak memory don't depend on the level	Measured
The curve flattens because each chunk is compressed on its own	Inferred from the format; chunk sizes not measured
Why level 13 and levels 2–4 come out larger	Not investigated
The same results for practice, qualifying and time-trial sessions	Not measured; only a race was tested
Live recording cost, as opposed to offline conversion	Not measured; the extra time above lands on the writer thread during a session

7. Options

Ordered by saving for the effort. None of them drops data. **Saving** is for the 92-minute race's all-drivers file.

#	Option	Saving	Cost and risk
1	Keep level 9	—	None
2	Raise the live writer to level 15	1.7 MB (2.2%), measured	About +80 s of writer CPU per race. Low risk
3	Re-encode a finished file at level 18–20 after the session ends	2.5–2.8 MB (3.3–3.6%), measured as conversion	New code; several minutes of background CPU per race; the file is replaced after it's written
4	Compress larger units (several laps, or a shared dictionary per data type)	Unknown, not measured	Format change; affects seeking and the memory model

8. Observations

Overall: keep level 9. It sits at the knee of the curve. Below it, files grow by up to 5% for no gain in speed. Above it, a few percent costs minutes per race.

- **The level isn't where the size is.** All 22 levels land within 9% of each other. Anything bigger would have to come from what goes into each frame (option 4), not from how hard each frame is squeezed.
- **Level 15 is the only cheap step up**, and it's still only 1.7 MB a race.
- **Driver-only recordings are already small:** 4.9 MB a race at level 9. The best any level does there is 0.18 MB less.

Appendix: all levels

Level	All drivers (MB)		vs 9	Driver only (MB)		vs 9	Time (s)	vs 9	Open all / driver (s)	Peak (MB)
1	79.36		+3.8%	5.16		+4.8%	183	1.0×	1.49 / 1.17	355
2	79.73		+4.3%	5.18		+5.3%	139	0.7×	1.41 / 1.37	353
3	80.12		+4.8%	5.17		+5.1%	176	0.9×	1.67 / 1.11	357
4	80.01		+4.7%	5.18		+5.2%	186	1.0×	1.46 / 1.10	357
5	78.71		+3.0%	5.06		+2.9%	160	0.9×	1.35 / 1.10	348
6	77.63		+1.6%	4.98		+1.3%	151	0.8×	1.39 / 1.07	365
7	77.24		+1.0%	4.95		+0.6%	178	1.0×	1.37 / 1.09	349
8	76.95		+0.7%	4.93		+0.1%	183	1.0×	1.44 / 1.10	350
9	76.44		0.0%	4.92		0.0%	187	1.0×	1.41 / 1.08	351
10	76.37		−0.1%	4.91		−0.2%	191	1.0×	1.69 / 1.08	360
11	76.40		0.0%	4.92		−0.1%	190	1.0×	1.38 / 1.08	350
12	76.34		−0.1%	4.90		−0.4%	202	1.1×	1.42 / 1.09	354
13	78.09		+2.2%	4.96		+0.7%	215	1.2×	1.35 / 1.08	365
14	76.38		−0.1%	4.87		−1.0%	169	0.9×	1.36 / 1.12	354
15	74.74		−2.2%	4.80		−2.4%	270	1.4×	1.40 / 1.08	357
16	74.84		−2.1%	4.80		−2.4%	301	1.6×	1.38 / 1.11	349
17	74.13		−3.0%	4.77		−3.1%	498	2.7×	1.41 / 1.12	363
18	73.90		−3.3%	4.76		−3.4%	544	2.9×	1.33 / 1.07	343
19	73.90		−3.3%	4.76		−3.3%	743	4.0×	1.79 / 1.43	355
20	73.68		−3.6%	4.75		−3.5%	1023	5.5×	1.42 / 1.11	347
21	73.69		−3.6%	4.75		−3.5%	972	5.2×	1.42 / 1.11	393
22	73.60		−3.7%	4.74		−3.6%	1370	7.3×	1.42 / 1.10	396